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RESEARCH

# From Efficiency to Equity: Assessing Supply–Demand Fairness in Underground Pedestrian System Safety in Japanese Cities

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## Abstract

Underground Pedestrian Systems (UPSs) are important components of urban resilience, yet existing safety assessments often remain efficiency-oriented and insufficiently address equity and vulnerability. This study develops an integrated supply-demand-equity framework to evaluate UPS safety from a disaster risk reduction perspective. Combining geospatial, socio-demographic, and infrastructural data, the framework constructs the Safety-support Supply Index (SSI), Safety Demand Index (SDI), Supply-Demand Mismatch Index (SDMI), Socially Weighted Demand Equity Index (SDEI), and Group Equity Index (GEI). Four typologically contrasting Japanese UPSs, namely Tokyo, Shinjuku, Sapporo, and Tenjin, are examined to diagnose network-based safety-support adequacy and distributive fairness. The results reveal pronounced spatial and social heterogeneity. Sapporo shows the highest relative supply adequacy and composite equity performance (GEI = 1.071), whereas Tenjin exhibits combined deficits in safety-support supply and equity performance (GEI = 0.453). Shinjuku shows a potential distributional equity deficit despite moderate safety-support capacity, while Tokyo presents a near-threshold equity profile. These contrasts suggest that built-environment conditions and governance arrangements may be associated with different safety-equity outcomes across UPS types, although the findings should be interpreted as exploratory comparative evidence rather than causal or statistically generalizable results. The proposed framework translates spatial justice into operational diagnostic indicators and provides a transparent tool for identifying supply–demand mismatch and group-sensitive safety deficits in subterranean urban systems. It is intended as an exploratory spatial-equity diagnostic framework rather than a hazard-specific engineering safety audit.

**Keywords** Underground Pedestrian Systems (UPS) · Supply-demand equity · Spatial equity · Infrastructure resilience · Japan

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**Abbreviations**

| | |
|---|---|
| UPS | Underground pedestrian system |
| SSI | Safety-support supply index |
| SDI | Safety demand index |
| SDMI | Supply–demand mismatch index |
| SDEI | Socially-weighted demand equity index |
| GEI | Group equity index |

## Introduction

As global urbanization accelerates, cities are increasingly looking to vertical and underground growth strategies. Consequently, Underground Pedestrian Systems (UPSs) have evolved from traditional auxiliary access routes into critical infrastructure at the heart of high-density environments (Dong et al., 2023). These networks are renowned for enhancing connectivity and business activity under all weather conditions (Cui, 2021). Simultaneously, they mitigate surface-level challenges, such as traffic congestion, land scarcity (He et al., 2016), and climate change impacts (He et al., 2024; Javanroodi et al., 2023). In an era of rising geopolitical tensions, public health crises, and frequent extreme weather events, UPSs have transcended their role as conveniences to become essential protected urban infrastructure. As the use of underground space intensifies, UPSs increasingly function as important components of urban mobility, emergency access, and climate-adaptive pedestrian infrastructure. Their safety and reliability are therefore not only matters of everyday convenience, but also important concerns for urban resilience and risk governance (Tsui et al., 2021; Zhao et al., 2021). However, the intensification of underground space usage introduces unprecedented risks (Kulikova & Balovtsev, 2020). Due to enclosed spaces, limited access points, and poor visibility, UPSs are highly susceptible to catastrophic failure during emergencies such as fires, earthquakes, or flooding (Wu et al., 2018). Ensuring the robustness and safety of these subterranean arteries has therefore become a paramount challenge in modern urban governance (Li & Wang, 2025; Liu et al., 2021).

However, the growing functional complexity of UPSs also introduces safety challenges. Enclosed spatial conditions, limited access points, vertical circulation barriers, and complex route structures may increase evacuation difficulty during emergencies such as fires, earthquakes, flooding, or extreme weather events. Existing studies on UPS safety have made important advances in structural safety, evacuation modelling, capacity optimization, and network robustness (Chen et al., 2023; Hou et al., 2022). Nevertheless, many assessments remain primarily efficiency-oriented, focusing on throughput, response time, evacuation rate, or aggregate accessibility (Lu et al., 2022; Xu & Xu, 2024). Such approaches often treat users as homogeneous pedestrians (Correa et al., 2024), while insufficiently accounting for differences in mobility capacity, cognitive ability, age, and vulnerability(Hou et al., 2022). As a result, the question of who can effectively access safety-support resources, and whether vulnerable groups are adequately protected, remains insufficiently addressed (Lu et al., 2023).

This gap is not merely a technical issue, but also an equity issue in urban infrastructure provision. From the perspective of infrastructure justice, safety resources are not distributed in socially neutral ways. The spatial configuration of entrances, passages, vertical links, refuge opportunities, and surface connections can produce uneven access to safety-support capacity across locations and user groups (Burgers, 2002; Coutard, 2008). In underground environments, these inequalities may be particularly pronounced because vertical depth, route complexity, and limited alternatives can disproportionately affect elderly users, children, persons with disabilities, and other mobility-constrained groups. Therefore, UPS safety assessment needs to move beyond aggregate efficiency and incorporate the distributive fairness of safety-support resources.

To address this issue, this study develops a supply-demand-equity framework for assessing UPS safety from an equity-oriented perspective. In this framework, safety-support supply is defined as network-based safety-support capacity, including access, egress, redundancy, and spatial robustness, rather than comprehensive hazard-specific engineering safety compliance. Safety demand refers to socially and spatially differentiated pressure generated by population vulnerability, activity concentration, and built-environment risk conditions. Equity is evaluated by examining whether safety-support capacity is distributed in a way that responds to differentiated social needs. The proposed framework is intended to complement, rather than replace, engineering evacuation models, fire-safety audits, flood-control assessments, or other hazard-specific safety evaluations.

Compared with conventional UPS studies that emphasize evacuation efficiency, accessibility performance, or network resilience, this study advances an equity-oriented diagnostic perspective in three ways. First, it reframes UPS safety provision as a supply-demand relationship, in which network-based safety-support capacity is assessed against socially and spatially differentiated demand. Second, it distinguishes baseline safety demand from socially weighted demand, thereby incorporating vulnerable groups into the assessment of safety need. Third, it integrates supply adequacy and distributional fairness through equity-sensitive indicators, translating the abstract concept of spatial justice into measurable diagnostic metrics. Rather than providing a definitive causal model, the framework offers an exploratory tool for identifying potential safety-equity deficits in underground pedestrian systems.

Accordingly, this study addresses the following research questions:

**RQ1** How can an integrated framework be constructed to assess supply-demand balance in UPS safety from an equity perspective?

**RQ2** What spatial patterns characterize the mismatch between network-based safety-support supply and socially differentiated demand across different UPS cases?

**RQ3** What contextual conditions and governance arrangements may be associated with these supply–demand mismatches?

**RQ4** How can the findings inform more inclusive and equity-sensitive risk governance strategies for underground pedestrian systems?

By addressing these questions, this study makes three contributions. First, it proposes a replicable geospatial diagnostic framework for assessing safety-equity issues in underground pedestrian systems. Second, it operationalizes a set of equity-sensitive indicators, including Safety-support Supply Index (SSI), Safety Demand Index (SDI), Supply-Demand Mismatch Index(SDMI), Socially-Weighted Demand Equity Index(SDEI) and Group Equity Index (GEI), to link network-based safety-support capacity with socially differentiated demand. Third, through a typological comparison of four Japanese UPS cases, it provides exploratory evidence on how different spatial configurations and governance contexts may be associated with divergent safety-equity outcomes. Overall, the study contributes to bridging infrastructure resilience and urban justice debates in subterranean urban environments.

## Study Area and Data Resources

### Study Area

Japan was selected as the empirical context for this study because it combines mature underground space development, high-density rail-oriented urbanization, recurrent exposure to natural hazards, and relatively developed infrastructure governance systems. Japanese cities have long developed underground pedestrian systems that are closely integrated with railway stations, commercial facilities, and surrounding surface pedestrian networks (Tsui et al., 2021; Yuan et al., 2019). This makes Japan a suitable context for examining how network-based safety-support capacity and socially differentiated safety demand interact in complex subterranean environments (Inamochi et al., 2021).

Three considerations support the selection of Japan. First, the country's compact urban form, limited land availability (Wan et al., 2023; Yuan et al., 2020), and rail-oriented development have encouraged the formation of highly integrated underground pedestrian networks around major stations (Ke et al., 2021; Liao & Scheuer, 2022). Second, Japanese cities are exposed to multiple hazards, including earthquakes, heavy rainfall, flooding, and extreme weather, which makes the safety and resilience of underground pedestrian environments an important urban governance issue (Zhu & Kan, 2012). Third, Japan's long-standing infrastructure planning and management systems provide a meaningful context for exploring how different spatial configurations and governance arrangements may be associated with safety-equity outcomes.

This study adopts a comparative typological case-study design rather than a statistically representative sampling strategy. The aim is not to represent all Japanese UPSs, but to compare contrasting types of underground pedestrian systems in terms of spatial morphology, functional role, climatic and risk context, and governance or development setting. Based on this logic, four typologically contrasting UPS cases were selected: Tokyo, Shinjuku, Sapporo, and Tenjin (Fig. 1).

Tokyo Station represents a nationally significant, vertically layered UPS embedded in a large-scale metropolitan transport and commercial complex (Inamochi et al., 2021; Zacharias et al., 2011). Shinjuku Station represents a high-density intermodal hub with intense passenger flow and complex transfer conditions (Fukuda




**Fig. 1** Spatial distribution of administrative divisions and major rail lines

et al., 2024; Yang & Yao, 2019). Sapporo Station represents a cold-climate UPS in which underground pedestrian space plays an important role in weather-protected movement and winter pedestrian continuity (Kishii, 2016; Krivorotko et al., 2024). Tenjin Station represents a decentralized regional commercial underground network in a compact urban context (Choi and Yoo, 2006). Together, these cases provide a comparative basis for diagnosing safety-equity patterns across several contrasting UPS types, rather than a nationally representative sample of Japanese UPSs (Fig. 2).

Table 1 further summarizes the case-selection logic. Tokyo is used as a large-scale benchmark case; Shinjuku as a high-demand stress case; Sapporo as a surplus and redundancy-oriented case; and Tenjin as a potential supply-deficit case. This typological contrast supports exploratory comparison while delimiting the scope of inference.

Potential sampling bias should nevertheless be acknowledged. The four cases are all located in major metropolitan centers and overrepresent relatively mature UPSs with substantial transport and commercial functions. Smaller, emerging, or less institutionally developed underground pedestrian systems may exhibit different safety-equity dynamics. Therefore, the findings should be interpreted as exploratory typological evidence rather than as nationally generalizable results for all Japanese UPSs.

The spatial extent of each case was not limited to the physical boundary of the underground facility. Because UPS safety demand is shaped by surrounding pedestrian flows, surface-level risk conditions, and the distribution of vulnerable populations, a 15-minute pedestrian catchment was delineated as the main analytical unit. The 15-minute threshold was adopted because it is commonly used in station-area, TOD, and pedestrian catchment studies to approximate the effective walking service

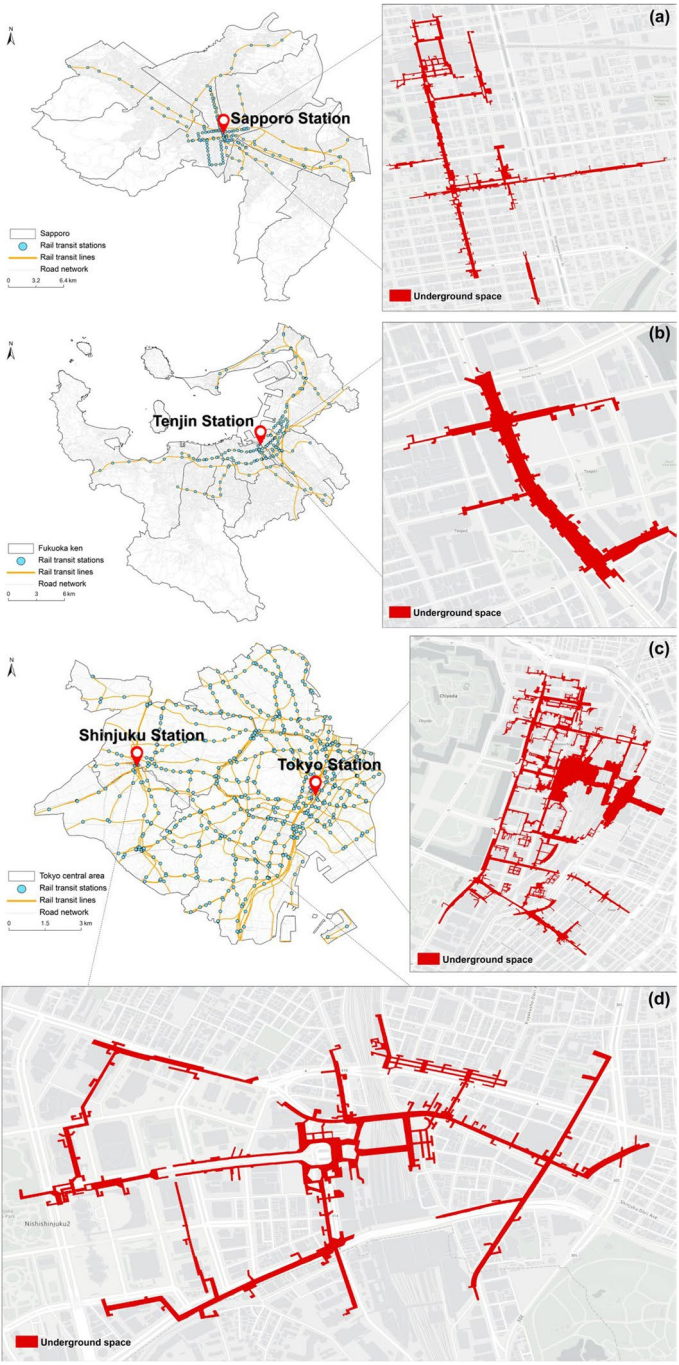



**Fig. 2** UPSs of the four stations: (**a**) Sapporo, (**b**) Tenjin, (**c**) Shinjuku, and (**d**) Tokyo

**Table 1** Case selection logic and typological contrast

| Case | Spatial morphology | Functional role | Climatic and risk context | Governance or development context | Analytical purpose |
|---|---|---|---|---|---|
| Tokyo | Vertically layered and complex | National gateway and metropolitan hub | High-density metropolitan context | State-corporate coordinated development context | Large-scale benchmark case |
| Shinjuku | Dense intermodal network | Mega transfer hub | High passenger-flow pressure | Commercially intensive and fragmented development context | High-demand stress case |
| Sapporo | Linear or binuclear UPS | Cold-climate pedestrian protection | Snow and winter climate adaptation | Public-planning and redundancy-oriented context | Surplus and redundancy case |
| Tenjin | Decentralized commercial network | Regional commercial core | Rainfall or flood exposure and compact urban form | Fragmented and market-responsive development context | Potential supply-deficit case |

area of transit-related facilities. Compared with Euclidean buffers, network-based pedestrian isochrones better reflect actual walking accessibility and the functional service area of station-related underground systems.

To examine whether the diagnostic results are sensitive to catchment delineation, a boundary sensitivity test was conducted using 10-, 15-, and 20-minute walking catchments. The 15-minute catchment was used as the baseline setting. For each alternative catchment, demand-side indicators were recalculated, while the network-based safety-support supply index was kept constant because SSI is defined by the internal UPS structure rather than the surrounding pedestrian catchment. The resulting SDMI values were then compared across the three catchment thresholds. The results, reported in Appendix E, Table E1, indicate that the main diagnostic typology remains stable, suggesting that the supply–demand mismatch interpretation is not merely an artefact of the selected 15-minute boundary.

## Data Sources and Processing

To operationalize the supply-demand-equity framework, this study constructed a multi-source geospatial database covering underground network structure, surface pedestrian networks, built environment characteristics, socio-demographic conditions, activity intensity, and transport demand. All spatial datasets were first harmonized in WGS 1984 for data integration and then projected into appropriate metric coordinate systems for distance, area, and network-based calculations.

The supply-side data describe the spatial-network characteristics of UPSs, including underground passages, entrances and exits, vertical circulation elements, passage length, and transfer complexity. These data were derived from official station maps, railway operator maps, Google Maps indoor or pedestrian references, and manual GIS digitization. The purpose of these data is to measure network-based safety-support capacity, including access, egress, redundancy, and spatial robustness, rather than comprehensive engineering safety compliance.

The demand-side data capture population vulnerability, activity concentration, and built-environment risk conditions within the walking catchments. Age-structured gridded population data were used to identify vulnerable population groups, while POIs, station ridership, building form, land use, and surface pedestrian networks were used to characterize functional intensity and contextual demand pressure. Because several datasets were derived from secondary or globally available sources, the resulting indicators should be interpreted as comparative measures rather than precise engineering measurements. Detailed datasets and sources are summarized in Table 2.

Because 3D-GloBFP is a globally derived building-form dataset, its height and footprint estimates may contain local uncertainties in dense urban environments. Therefore, it was used only to characterize relative built-environment conditions across cases, rather than to provide ground-truth building-height measurements.

**Table 2** Data sources, processing steps, and verification procedures

| Data category | Dataset | Source | Spatial and temporal resolution | Processing | Verification |
|---|---|---|---|---|---|
| UPS layout | Underground passages, entrances, exits, vertical links | Official station maps; railway operator maps; Google Maps indoor or pedestrian reference layers | Station level | Manual digitization and georeferencing | Cross-checked with official maps and street-view references |
| UPS passage attributes | Width, length, depth, transfer distance | Official station plans; operator websites; Google Maps walking paths | Segment and entrance level | GIS measurement and manual calibration | Checked against documented transfer distance or walking time where available |
| Surface network | Surface pedestrian and road network | Fundamental Geospatial Data of Japan; GSI basemap; OSM | Vector network | Topology cleaning and network construction | Compared with GSI basemap and OSM |
| Building form | Building footprint and height | 3D-GloBFP (Che et al., 2024) | Building-level; raster or vector | Clipped to walking catchments | Used for relative comparison; visually checked for abnormal values |
| Population | Age-structured gridded population | National Census of Japan | 250 m×250 m grid | Spatial join to walking catchments | Checked against census totals |
| POIs | Commercial, transit, and public facilities | GSI; local government open data portals | Point data | Classification, cleaning, and KDE | Duplicate removal and category validation |
| Ridership | Daily passenger flow | Annual statistical reports from railway operators | Station-level annual and daily average | Normalization and station matching | Cross-checked across operator reports where available |

## Conceptual Framework and Methodology

### Conceptualizing Supply, Demand and Equity in UPS Safety

#### Network-Based Safety-Support Supply

Safety-support supply refers to the capacity of a UPS to support access, egress, route redundancy, and spatial robustness under both ordinary and disrupted conditions. It is conceptualized as a form of network-based safety-support capacity rather than a complete engineering safety audit. This distinction is important because several hazard-specific safety components, including fire suppression systems, waterproof doors, drainage facilities, emergency power, smoke extraction, ventilation, and structural seismic resistance, are not directly measured in the SSI. Accordingly, SSI should be interpreted as an evacuation-supportive and accessibility-related spatial index.

Framing UPSs as socio-technical systems (Lei et al., 2025; Xie et al., 2025), this study decomposes safety-support supply into four interrelated dimensions. The first dimension is spatial coverage, which reflects the extent to which the UPS can provide safety-support access to surrounding buildings, pedestrian areas, and functional spaces. This dimension is linked to accessibility theory in urban geography, where access to safety-related infrastructure is understood as a critical public good (Kelobonye et al., 2020; Merlin & Jehle, 2023).

The second dimension is path redundancy, which captures the availability of alternative egress routes. Drawing on infrastructure resilience and network science (Xu et al., 2024; Zhao et al., 2021), redundancy can reduce dependence on single paths and improve the capacity of the system to absorb local disruptions (Raicu et al., 2019).

The third dimension is vertical accessibility, which reflects the degree to which underground, surface, and vertical circulation spaces are connected within a three-dimensional pedestrian network. The fourth dimension is network resilience, which is assessed through graph-theoretical measures such as betweenness centrality concentration and flow centralization (Pearce et al., 2021). These indicators help identify whether the UPS depends heavily on a small number of critical paths, which may increase vulnerability to cascading disruption (Tariverdi et al., 2023).

#### Safety Demand: a Multi-Dimensional Vulnerability Perspective

Safety demand is defined as a socially and spatially constructed expression of vulnerability. It is not equivalent to population density alone. Rather, demand emerges from the interaction among socio-demographic vulnerability, spatial-functional concentration, and built-environment risk conditions.

The first component is socio-demographic vulnerability. Social vulnerability theory suggests that different groups have unequal capacities to perceive, respond to, and recover from emergency situations (Cornu et al., 2024; Louzeiro et al., 2023). Older adults, children, adolescents, and persons with mobility constraints may face greater evacuation difficulty due to slower walking speed, lower physical endurance, reduced spatial cognition, or higher dependence on assistance (Tate et al., 2021; Tran

et al., 2020). Therefore, the concentration of vulnerable populations increases the demand for accessible and redundant safety-support resources.

The second component is spatial-functional concentration. High-density commercial districts, transport hubs, and mixed-use station areas generate intensive pedestrian flows and activity concentration (Liu et al. 2023a, b; Zhang et al. 2020). In emergencies, these areas may experience greater evacuation pressure because large numbers of users must access, navigate, or exit underground environments within a limited time window (Sung et al., 2022; Zhang et al., 2021).

The third component is built-environment risk context. Dense urban cores, high building density, limited open spaces, and weak surface pedestrian connectivity can increase the pressure placed on underground pedestrian infrastructure (Chen, 2021; Del Pinto et al., 2024; Helderop & Grubesic, 2019; Lall & Deichmann, 2012). In this study, built-environment risk is interpreted as a contextual condition co-varying with station location, UPS morphology, and design complexity, rather than as an independent causal determinant of safety-support capacity (Xu et al., 2024). This cautious interpretation responds to the limitation that the four-case typological design cannot support causal inference (Jin et al., 2021; Wei et al., 2020).

It should also be noted that safety demand in this study is operationalized primarily as a static, average-day structural condition, based on available census, land use, POI, built-environment, and daily ridership data. This approach does not directly capture dynamic demand fluctuations such as peak-hour surges, seasonal events, weather-related crowding, or emergency-induced behavioral changes.

### From Supply-Demand Mismatch to Safety Equity

The relationship between safety-support supply and safety demand provides the theoretical basis for diagnosing equity in UPS safety. A mismatch between high demand and low supply indicates not only a technical insufficiency, but also a potential distributive problem. Drawing on the perspectives of spatial justice and socio-technical infrastructure systems (Fainstein, 2014; Graham & Marvin, 2002; Soja, 2010), this study interprets UPS safety as a public infrastructure issue in which access to safety-support resources should be responsive to differentiated social needs (Grimley et al., 2024; Hosseini et al., 2022; Lu & Chen, 2024).

Two forms of inequity are emphasized. The first is spatial inequity, where safety-support capacity is unevenly distributed across station areas and may not correspond to areas of higher demand. The second is group-level inequity, where the same physical infrastructure may be experienced differently by different user groups (Ashik et al., 2020; Guo et al., 2019). In underground spaces, vertical depth, long walking distances, route complexity, and insufficient barrier-free facilities may disproportionately affect older adults, children, and mobility-constrained users (Sharma & Patil, 2024; Wan & Titheridge, 2024). These conditions can generate constructed vulnerability, where the built environment itself contributes to unequal access to safety-support resources. Figure 3 presents the conceptual framework of the supply-demand-equity relationship in UPS safety. The framework links network-based safety-support supply, socially and spatially differentiated demand, and equity-sensitive evaluation.

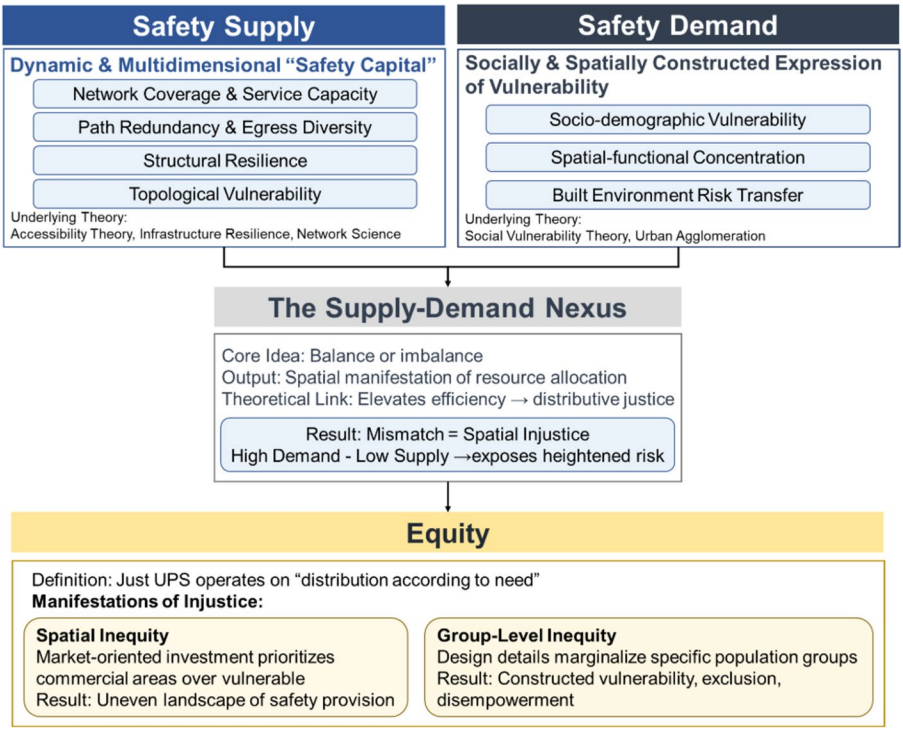



**Fig. 3** The supply–demand–equity relationship in UPS safety

## Analytical Framework and Indicator System

### Analytical Workflow and Index Relationships

The analytical framework consists of three stages (Fig. 4). The first stage is data integration and network modelling, in which underground network topology, entrance

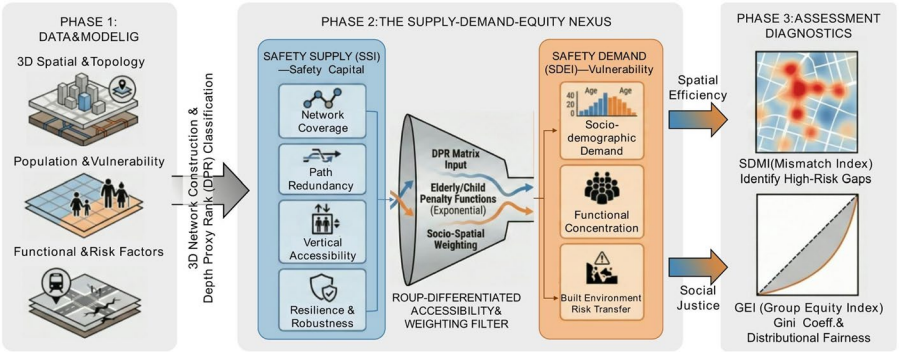



**Fig. 4** Research framework

and exit locations, vertical links, surface pedestrian networks, built environment data, socio-demographic data, POIs, and ridership data are integrated into a geospatial database. The second stage is supply and demand quantification, in which SSI and SDI are constructed to represent network-based safety-support supply and baseline safety demand. The third stage is mismatch and equity diagnosis, in which SDMI, SDEI, group-adjusted safety supply, and GEI are calculated.

The relationship among the core indices is summarized in Table 3. SSI measures network-based safety-support capacity. SDI measures baseline structural demand. SDMI evaluates the relative adequacy of supply against demand. SDEI introduces a normative adjustment to demand by prioritizing vulnerable groups. GEI integrates distributional fairness and needs-responsive adequacy to diagnose group-sensitive safety equity. This sequence clarifies that the framework combines descriptive empirical measurement with normative equity evaluation, rather than treating all indices as equivalent.

### Indicator System

To operationalize the framework, this study constructs a multi-dimensional indicator system covering safety-support supply, safety demand, mismatch, and group-level equity. The indicator set synthesizes recent advances in spatial service capacity (Liu et al. 2023a, b), network analysis (Aravind et al., 2024), spatial accessibility modelling (Dong et al., 2021), and equity diagnostics in the urban built environment (Li et al., 2023), and is detailed in Table 4.

The supply-side indicators capture the spatial-network characteristics of UPSs. These include service-area coverage, egress path redundancy, three-dimensional accessibility, and network resilience. The demand-side indicators capture socio-

**Table 3** Analytical roles and relationships among SSI, SDI, SDMI, SDEI, and GEI

| Index | Full name | Analytical role | Interpretation |
|---|---|---|---|
| SSI | Safety-support Supply Index | Network-based safety-support capacity | Higher SSI indicates stronger spatial-network support for access, egress, redundancy, and robustness |
| SDI | Safety Demand Index | Baseline structural safety demand | Higher SDI indicates stronger average-day structural demand pressure |
| SDMI | Supply-Demand Mismatch Index | Relative adequacy | SSI/SDI; values above 1 indicate relative surplus, whereas values below 1 indicate deficit |
| SDEI | Socially Weighted Demand Equity Index | Equity-sensitive demand | Adjusts demand by prioritizing vulnerable groups |
| GEI | Group Equity Index | Composite equity diagnosis | Integrates distributional fairness and needs-responsive adequacy |

**Table 4** Indicators for UPS safety-support supply, demand, mismatch, and equity assessment

| Primary dimension | Secondary dimension | Indicator | Definition and calculation method |
|---|---|---|---|
| Safety-support supply | Spatial coverage | Service area | Network service-area modelling; calculates built-up area covered by the UPS within 50 m, 100 m, 200 m, and 400 m evacuation-support radii |
| | Path redundancy | Number and total length of egress paths | Counts alternative network paths from building points to UPS entrances or exits using network analysis |
| | Vertical accessibility | Comprehensive 3D accessibility | Measures integrated accessibility of the combined surface, underground, and vertical pedestrian network |
| | Network resilience | Path betweenness centrality concentration index | Measures dependency on critical paths and potential vulnerability to single-point disruption |
| Safety demand | Socio-demographic demand | Vulnerable population ratio | Proportion of population aged $\geq 60$ years and $\leq 18$ years from age-structured population grids |
| | | Population density | Population load per unit of building or catchment area |
| | Spatial-functional demand | Public activity agglomeration | Kernel density of transport, commercial, and public-service POIs |
| | | Passenger flow concentration | Daily passenger ridership intensity at associated railway or metro stations |
| | Built-environment risk context | Building density | Density of building footprints or built-up intensity within the catchment |
| | | Street network vulnerability | Composite indicator reflecting surface road density and intersection spacing; lower connectivity indicates greater surface evacuation constraint |
| | | Surface refuge capacity | Number or density of surface refuge points; lower capacity indicates greater dependence on UPS safety-support functions |
| Mismatch and equity | Supply-demand mismatch | SDMI | Ratio of SSI to SDI after standardization and weighting |
| | Group-level distribution | Group-adjusted Gini coefficient | Gini coefficient of group-adjusted safety supply |
| | Socially weighted demand | SDEI | Demand index adjusted by vulnerable-group priority |
| | Composite equity | GEI | Composite index integrating distributional fairness and needs-responsive adequacy |

demographic vulnerability, spatial-functional concentration, and built-environment risk context. The mismatch and equity indicators evaluate whether supply is adequate relative to demand and whether safety-support resources are equitably accessible to different user groups.

## Evaluation Methods

### Safety-Support Supply and Demand Indices

Min-Max normalization is applied to all raw variables for the SSI and SDI to ensure comparability and robustness. This process scales all variables to a [0, 1] range,

effectively eliminating issues arising from different units and inconsistent directions. Indicators are categorized as either positive, where higher values signify better conditions (e.g., spatial service coverage), or negative, where higher values denote worse conditions (e.g., the proportion of elderly residents or fragility indices). To derive data-driven descriptive weights for SSI and SDI, the entropy weighting method is employed.

The entropy weighting method is used only for the construction of SSI and SDI, because these two indices aim to capture the empirical variation of network-based safety-support supply and baseline structural demand indicators across the selected UPS cases. In this sense, entropy weighting serves a descriptive purpose: indicators with greater cross-case variation are assigned higher information weights. It does not imply a normative judgment about which social group should be prioritized in safety provision. Because the entropy weights are derived from four typological cases, the resulting weights may be sensitive to individual cases. To address this limitation, we added two robustness-oriented classification-stability checks based on the available composite results. Appendix F, Table F2 reports recalculated SDMI and GEI values under equal-weight and dimension-balanced assumptions, while Appendix F, Table F3 reports a leave-one-out classification-stability check. These checks are used to assess the stability of the diagnostic classifications rather than to provide full statistical validation or population-level weight estimates.

This technique quantifies the degree of disorder or information contribution for each variable, consequently assigning higher weights to indicators exhibiting greater variability and discriminative power. Ultimately, the SSI and SDI are constructed as weighted composite scores, derived from these normalized values and their respective entropy-derived weights.

The Safety-support Supply Index is then calculated as:

$$SSI_i = \sum_{j=1}^{m_s} w_j^s \cdot x^s{}_{ij} \tag{1}$$

$$SDI_i = \sum_{k-1}^{m_d} w_k^d \cdot x^d{}_{ik} \tag{2}$$

where $m_s$ and $m_d$ are the number of supply and demand indicators, respectively; $w_j^s$ and $w_k^d$ are entropy weights for supply and demand indicators. The raw analytical outputs, normalization rules, entropy-weighting procedure, dimension-level aggregation logic, and SSI/SDI aggregation process are provided in Appendix A.

### Supply-Demand Mismatch Index

To evaluate the relative adequacy of safety-support supply against structural demand, the Supply–Demand Mismatch Index is calculated as:

$$SDMI_i = \frac{SSI_i}{SDI_i + \varepsilon} \tag{3}$$

Where $\epsilon$ is a small constant used to avoid division by zero. A value greater than 1 indicates that network-based safety-support supply exceeds baseline demand. A value close to 1 indicates approximate balance. A value below 1 indicates a potential deficit, where demand exceeds supply.

The SDMI should be interpreted as an indicator of chronic structural mismatch rather than a real-time congestion or evacuation-performance measure. It does not capture minute-by-minute crowding, behavioral response, weather-related fluctuation, or emergency-specific evacuation dynamics.

#### Descriptive Association Analysis

To explore how different categories of demand pressure align with safety-support supply, the demand-side indicators are grouped into three dimensions: socio-demographic demand, spatial-functional demand, and built-environment risk context. Dimension-level factor scores are calculated by averaging the normalized indicators within each group.

Pearson correlation coefficients are then calculated between SSI and the individual or grouped demand indicators. However, given the limited number of typological cases, these coefficients are used only as descriptive measures of association rather than inferential statistics. They do not support statistical significance testing, causal inference, or generalized variable-importance claims. Instead, they serve as exploratory indicators for identifying patterns that may warrant further testing in larger samples.

#### Demand Amplification Scenario Analysis

Because the baseline SDI captures average-day structural demand, it cannot represent dynamic fluctuations such as peak-hour surges, event-driven crowding, seasonal variation, or emergency-induced demand shifts. To address this limitation, a demand amplification scenario analysis is introduced.

The adjusted demand index under scenario $q$ is calculated as:

$$SDI_i^q = \lambda_q SDI_i \tag{4}$$

where $\lambda_q$ is the demand amplification coefficient, with $\lambda_q \in \{1.00, 1.25, 1.50, 2.00\}$.

These correspond to average-day conditions, moderate peak-hour stress, strong peak-hour stress, and extreme event or disruption-induced crowding. The scenario-based mismatch index is calculated as:

$$SDMI_i^q = \frac{SSI_i}{SDI_i^q + \varepsilon} \tag{5}$$

The purpose of this test is not to replace real-time pedestrian flow modelling, but to examine whether the diagnostic typology is sensitive to plausible increases in demand. The scenario results are reported in Appendix E, Table E2.

## Group-Weighted Equity Assessment

Following the supply–demand mismatch and scenario analysis above, this section proposes a proxy based group-weighted safety equity assessment framework, whose complete analytical workflow is depicted in Fig. 5.

### DPR as an Ordinal Proxy for Egress Difficulty

A key methodological challenge in evaluating group-differentiated safety accessibility is the absence of standardized entrance-level data on elevators, escalators, ramps, platform depth, indoor route geometry, and observed evacuation time. To address this data constraint, this study introduces the Depth Proxy Rank (DPR) as an ordinal proxy for relative egress difficulty.

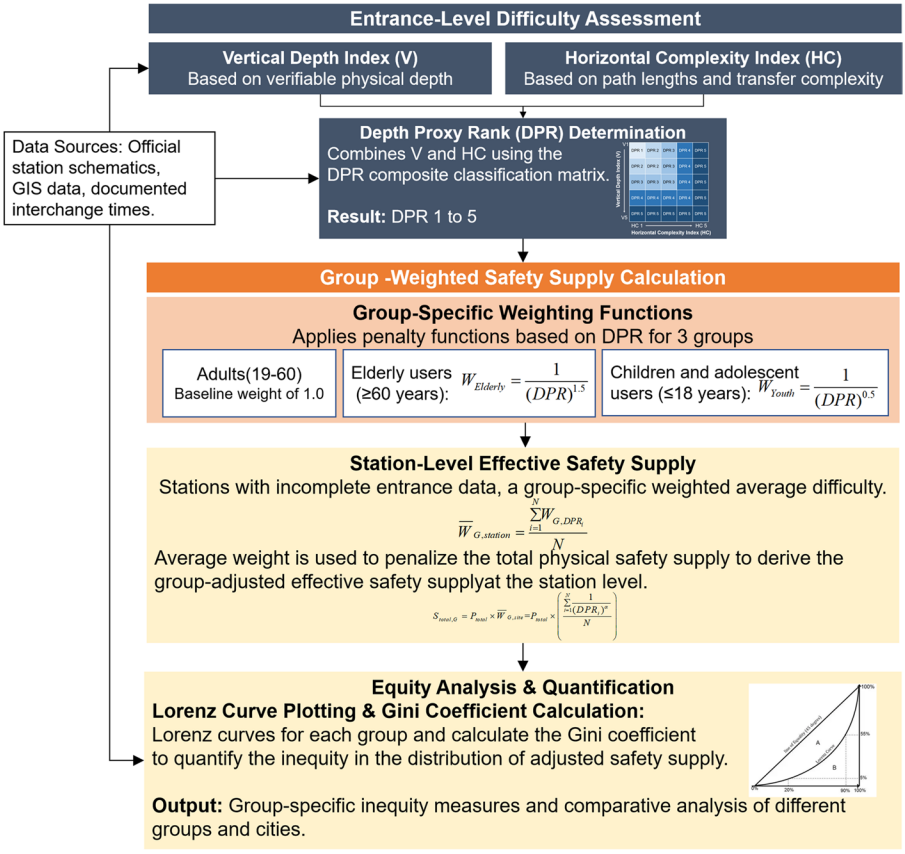



**Fig. 5** Workflow for proxy-based group-weighted safety equity assessment in UPSs

The DPR is designed as a comparative proxy for entrance-level egress difficulty under data constraints, rather than as a direct measure of barrier-free accessibility, actual evacuation time, or user-perceived difficulty. Its purpose is to support macro-level comparison across UPS cases where standardized data on elevators, escalators, ramps, platform depth, and detailed indoor geometry are incomplete or unavailable. Therefore, the DPR should be interpreted as an indicative ranking of relative egress difficulty, not as a substitute for on-site accessibility audits, evacuation experiments, or user perception surveys.

DPR is constructed through two operational dimensions. The first is the Vertical Depth Index (V), which classifies the depth of underground access or platform levels. The second is the Horizontal Complexity Index (HC), which classifies path length, transfer complexity, and horizontal navigation burden. The thresholds used in Tables 5 and 6 are operational classification rules rather than empirically validated physiological thresholds or regulatory standards. Their purpose is to support comparative ranking of relative egress difficulty, not to estimate exact evacuation time.

The final DPR value is determined by combining the V and HC scores through a composite classification matrix. This matrix-based approach ensures that both vertical depth and horizontal complexity are considered, but the resulting DPR should be interpreted as an algorithmic and reproducible proxy rather than a fully objective or behaviorally validated measure (Fig. 6).

To improve the transparency of DPR assignment, a three-step verification procedure is used. First, V-scores are checked against official station floor plans, public cross-section diagrams, and railway operator information. Second, HC-scores are compared with GIS-measured egress path lengths and documented transfer distances or walking times where

**Table 5** Operational classification criteria for vertical depth index (V)

| V score | Vertical difficulty | Operational classification criteria |
|---|---|---|
| V1 | Very Low | At-grade (street-level) or B1 (sub-level 1). |
| V2 | Low | B2 (sub-level 2). Typically standard cut-and-cover lines. |
| V3 | Moderate | B3 (sub-level 3). Typically medium-depth shield-tunnel lines. |
| V4 | High | B4 (sub-level 4). Deep shield-tunnel lines. |
| V5 | Very High | ≤B5 (sub-level 5 or deeper). Ultra-deep structures. |

**Table 6** Operational classification criteria for horizontal complexity index (HC)

| HC score | Horizontal complexity | Operational classification criteria |
|---|---|---|
| HC1 | Very Low | Single-purpose entrance (non-interchange) or path distance < 100 m. |
| HC2 | Low | Simple interchange (e.g., 2 lines) or path distance 100–200 m. |
| HC3 | Moderate | Standard interchange (> 2 lines) or path distance 200–350 m. |
| HC4 | High | Complex/long-distance interchange or path distance 350–500 m. |
| HC5 | Very High | Extreme/documented long walk interchange or path distance > 500 m. |

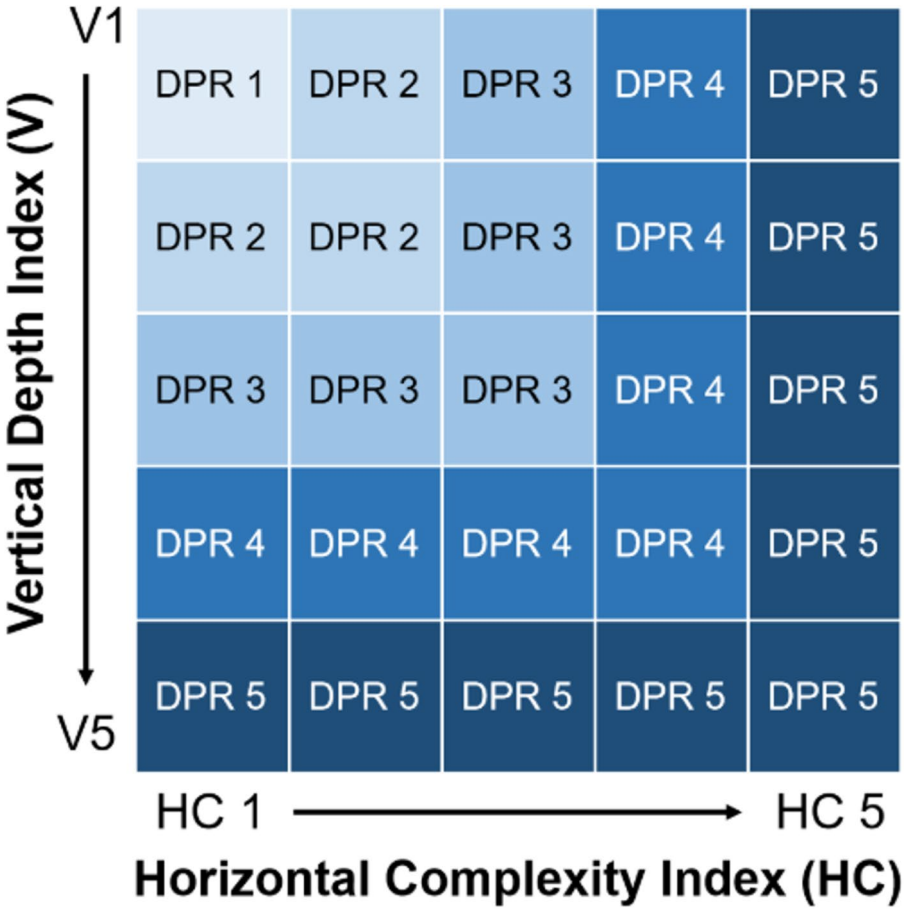



**Fig. 6** DPR composite classification matrix for proxy-based egress-difficulty ranking

available. Third, ambiguous cases are manually reviewed using station maps, indoor navigation information, and publicly available visual references. These procedures reduce the arbitrariness of DPR assignment, but they do not constitute full validation against observed evacuation behavior or user experience. Representative DPR classifications and verification evidence are reported in Appendix D, Tables D1 and D2. The sensitivity of GEI to alternative DPR-related assumptions is reported in Appendix F, Table F1.

### Group-Specific Adjustment of Effective Safety Supply

The purpose of group-specific adjustment is to estimate how the same network-based safety-support supply may become functionally less accessible to vulnerable groups under higher egress difficulty. This adjustment is normative rather than purely descriptive. Entropy weights describe empirical variation in SSI and SDI, whereas

group-specific weights introduce an equity-oriented correction based on differentiated mobility capacity.

Adults aged 19–60 years are used as the reference group:

$$w_{adult,i} = 1 \tag{6}$$

Older adults are assigned a stronger nonlinear penalty because environmental barriers may disproportionately increase physical effort, perceived difficulty, and evacuation time for this group (Bottomley & Lewis, 2020; Wagner et al., 2019; Rantakokko et al., 2016):

$$W_{Elderly} = \frac{1}{(DPR)^{1.5}} \tag{7}$$

Children and adolescents are assigned a milder nonlinear penalty:

$$W_{Youth} = \frac{1}{(DPR)^{0.5}} \tag{8}$$

where $DPR_i$ is the egress difficulty rank of entrance or segment i. These functional forms should be interpreted as normative assumptions for equity-sensitive diagnosis rather than empirically validated behavioral functions. Their influence is tested through sensitivity analysis.

For each station, the group-specific average accessibility weight is calculated as:

$$\overline{W}_{(G,station)} = \frac{\sum_{i=1}^{N} W_{G,DPR_i}}{N} \tag{9}$$

where N denotes the total number of entrances and $w_{G,i}$ is the group-specific weight derived from the DPR value of entrance $E_i$. The group-adjusted effective safety supply is then computed as:

$$S_{total,G} = P_{total} \times \overline{W}_{G,\,site} = P_{total} \times \left( \frac{\sum_{i=1}^{N} \frac{1}{(DPR_i)^a}}{N} \right) \tag{10}$$

This formulation estimates the relative reduction in effective safety-support supply experienced by different user groups. The resulting group-adjusted supply values are used to construct the Lorenz curve and group-specific Gini coefficient. Higher Gini values indicate greater inequality in access to effective safety-support resources.

### Composite Equity Evaluation for Vulnerable Groups

The SDI captures baseline structural demand but does not by itself express the normative principle that safety-support resources should prioritize vulnerable groups. To address this limitation, this study constructs the Socially Weighted Demand Equity Index (SDEI). The SDEI modifies baseline demand by increasing the weight assigned to vulnerable population exposure.

The SDEI is calculated as:

$$SDEI = (1 - \theta)SDI_s + \theta VUL_s \tag{11}$$

Where VULs is the standardized vulnerable population component, including older adults, children, and adolescents, and $\theta$ is the vulnerability priority coefficient. In the baseline setting, $\theta$=0.40. This means that vulnerable-group exposure accounts for 40% of the socially weighted demand index. This parameter is normative and should be interpreted as an equity-oriented weighting choice rather than a statistically estimated coefficient.

The Group Equity Index integrates two components, distributional fairness and needs-responsive adequacy. It is calculated as:

$$GEI_s = \alpha \times (1 - Gini_s) + \beta \times \left( \frac{SSI_s}{SDEI_s + \varepsilon} \right) \tag{12}$$

where $1-Gini_s$ represents distributional fairness, $SSI_s$ / $SDEI_s$ represents needs-responsive adequacy, and $\alpha+\beta=1$. In the baseline setting, $\alpha=0.5$ and $\beta=0.5$, giving equal importance to fairness and adequacy.

A GEI value greater than 1 indicates relatively favorable composite equity, while a value below 1 indicates a potential equity deficit. However, because GEI incorporates DPR-based group adjustment and normative weighting parameters, it should be interpreted as an indicative diagnostic estimate rather than a precise measurement of actual evacuation accessibility.

To examine whether GEI classifications are sensitive to normative weighting choices, GEI is recalculated under five alternative α–β combinations, ranging from fairness-oriented to adequacy-oriented schemes. The sensitivity of GEI to alternative normative weighting schemes is reported in Appendix F, Table F4. These tests are used to assess whether the main equity typologies remain stable and to identify cases that are sensitive to alternative normative assumptions.

## Results

### UPS Safety Supply and Demand Characteristics

#### Heterogeneity in Safety Supply

The SSI results reveal clear heterogeneity in network-based safety-support capacity across the four UPS cases (Fig. 7). The supporting service-area maps, path-redun-

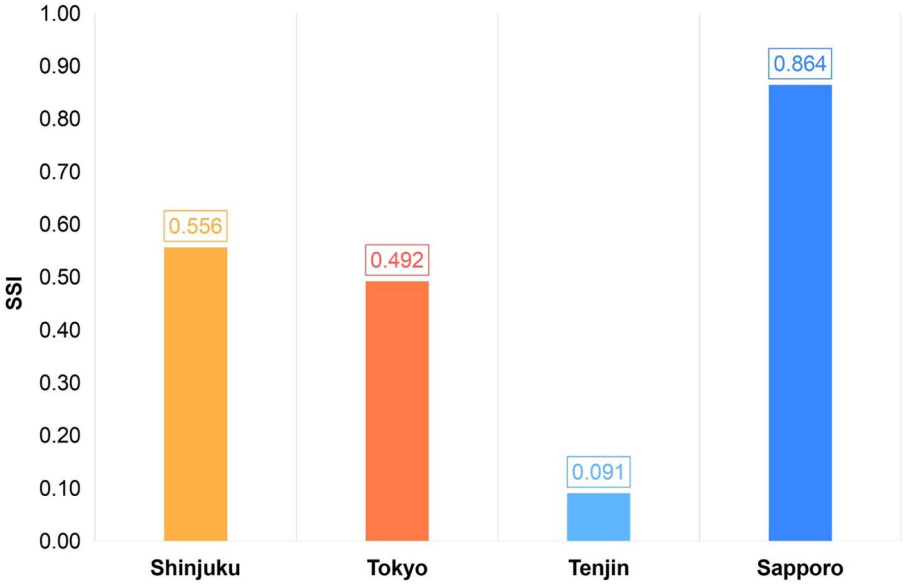



**Fig. 7** Comparative SSI scores for UPSs

dancy results, vertical-accessibility visualization, and TPBt-based network-resilience results are reported in Appendix B.

Sapporo records the highest SSI value (0.864), indicating comparatively strong evacuation-supportive spatial capacity. This advantage is mainly reflected in its route redundancy, interconnected network structure, and relatively balanced spatial coverage.

Shinjuku (SSI = 0.556) and Tokyo (SSI = 0.492) show moderate levels of safety-support supply. Their scores indicate that both large-scale hubs possess substantial network capacity, but also exhibit potential constraints related to vertical accessibility and the concentration of network stress on selected critical paths.

Tenjin records the lowest SSI value (0.091), indicating a marked deficit in network-based safety-support capacity. This low score is primarily associated with limited egress redundancy, weak spatial coverage, and insufficient vertical accessibility. Therefore, Tenjin's low SSI should be interpreted as evidence of limited network-based safety-support capacity, particularly in terms of egress redundancy, spatial coverage, and vertical accessibility, rather than as a direct assessment of engineering structural safety.

### Differentiated Profiles of Safety Demand

The SDI results indicate differentiated demand profiles across the four UPS cases, reflecting variations in socio-demographic vulnerability, functional agglomeration, and built-environment risk context (Fig. 8). Detailed SDI indicator definitions and calculation procedures are reported in Appendix C.

Shinjuku records the highest SDI value (0.704), suggesting the strongest structural demand pressure among the four cases. This result is associated with high popula-

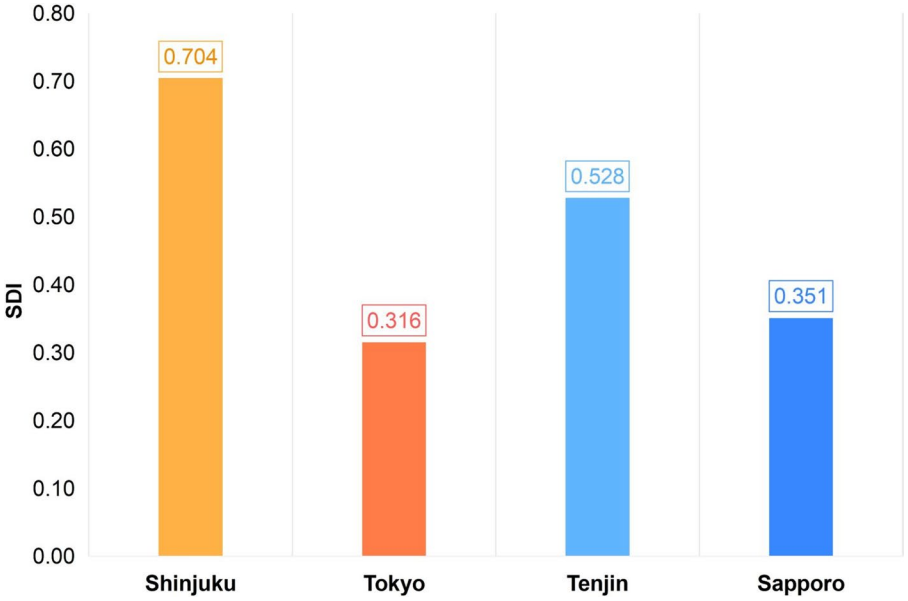



**Fig. 8** Comparative SDI scores for UPSs

tion density, intense ridership, concentrated commercial and transit functions, and a complex built environment. Together, these conditions generate substantial pressure on the surrounding underground pedestrian network.

Tenjin ranks second in demand intensity (SDI = 0.528). Although its demographic and ridership indicators are lower than those of Shinjuku, its role as a regional commercial core, compact urban form, and built-environment constraints contribute to a relatively high safety-demand burden.

Sapporo (SDI = 0.351) and Tokyo (SDI = 0.316) present lower baseline demand levels. Sapporo's lower SDI is related to relatively moderate population density and functional intensity within the catchment. Tokyo, despite its national gateway function, shows the lowest SDI in this analysis, reflecting the specific demographic and land-use composition of its immediate walking catchment.

### Exploratory Association Patterns

Building upon the SSI and SDI results, this section provides an exploratory comparison of how demand-side pressures are aligned with network-based safety-support capacity across the four typological cases. Given the small number of cases, the coefficients shown in Figs. 9 and 10 are not interpreted as statistically significant correlations or causal effects. Rather, they serve as descriptive indicators to support pattern recognition and hypothesis generation.

Built-environment risk should be interpreted as a contextual condition co-varying with UPS morphology and station location, rather than an independent causal factor. Because UPSs are embedded in dense station-area development, the observed asso-

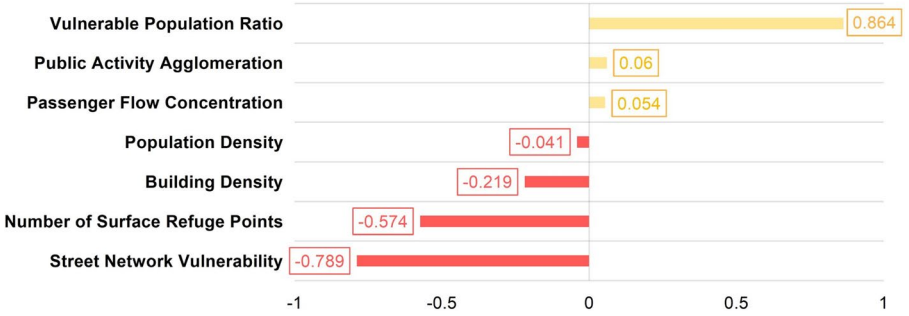



**Fig. 9** Descriptive association coefficients between demand indicators and SSI

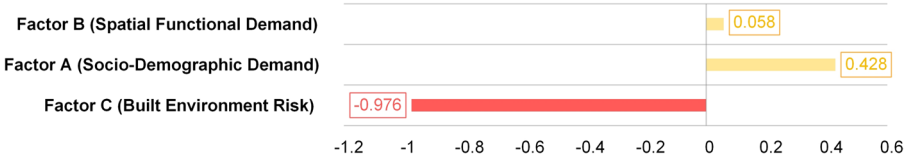



**Fig. 10** Descriptive association coefficients between aggregated demand factors and SSI

ciation may partly reflect co-location and design difficulty rather than a unidirectional effect of surface risk on underground safety-support capacity.

Coefficients are calculated for four typological cases and are used for exploratory description only; they are not interpreted as inferential statistics or causal effects.

Built Environment Risk (Factor C) shows a strong inverse descriptive association with SSI in the four-case comparison ($r=-0.976$). However, given the limited sample size, this coefficient should not be interpreted as statistically significant or causal. The pattern suggests that UPSs embedded in denser and more complex surface environments may face greater difficulty in achieving high levels of network redundancy, vertical accessibility, and spatial robustness. At the indicator level, street network vulnerability ($r=-0.789$) and building density ($r=-0.219$) both show negative descriptive associations with SSI.

In contrast, spatial functional demand (Factor B), which includes measures of commercial intensity and ridership, shows virtually no descriptive association with SSI (r=+0.058). This weak association suggests that, within the four selected cases, higher activity intensity is not always accompanied by proportionally higher network-based safety-support capacity. This finding should be interpreted as a hypothesis-generating observation rather than a general rule.

The role of socio-demographic demand (Factor A) is more nuanced. While the overall descriptive association with SSI is moderate (r=+0.428), this is largely attributable to the positive descriptive association between the proportion of vulnerable populations and SSI (r=+0.864). Population density, by contrast, exhibits no meaningful descriptive association ($r=-0.041$).

This pattern may reflect, in some contexts, the influence of safety-oriented planning or historical development conditions, although such interpretation remains tentative and requires larger-sample verification. An unexpected negative descrip-

tive association is observed between the number of surface-level refuge points and SSI ($r=-0.574$), suggesting that a higher provision of such facilities does not correspond with greater overall safety capacity. This apparent contradiction may reflect two dynamics. First, a higher density of refuge points may indicate areas already recognized as high-risk, thus confounding the association. Second, it may reveal the limitations of quantitative facility metrics: the mere presence or count of safety elements does not capture their spatial integration, operational quality, or effectiveness in real emergency scenarios.

In sum, the association results suggest that built-environment context may be linked to differences in network-based safety-support capacity across the four typological cases. However, these patterns should be interpreted as exploratory evidence rather than statistical proof. They provide a basis for discussion and future hypothesis testing, but do not establish built-environment or governance causality.

### Supply–Demand Mismatch Diagnosis

A comprehensive diagnosis of UPS safety-support performance requires the integration of both supply- and demand-side conditions. The SDMI was calculated for each case to evaluate the relative adequacy of network-based safety-support supply against baseline structural demand.

The SDMI values reveal substantial divergence among the four cases (Fig. 11). Under the baseline SDI formulation, Sapporo shows the highest relative supply adequacy (SDMI = 2.46), followed by Tokyo (SDMI = 1.56). In contrast, Shinjuku (SDMI = 0.79) and Tenjin (SDMI = 0.17) fall below the adequacy threshold, indicating potential supply–demand deficits. Tenjin exhibits the most pronounced mismatch, reflecting the combination of low SSI and relatively high demand pressure.

To support typological interpretation, a four-quadrant diagnostic model was developed (Fig. 12). The model positions each UPS in a two-dimensional analytical space

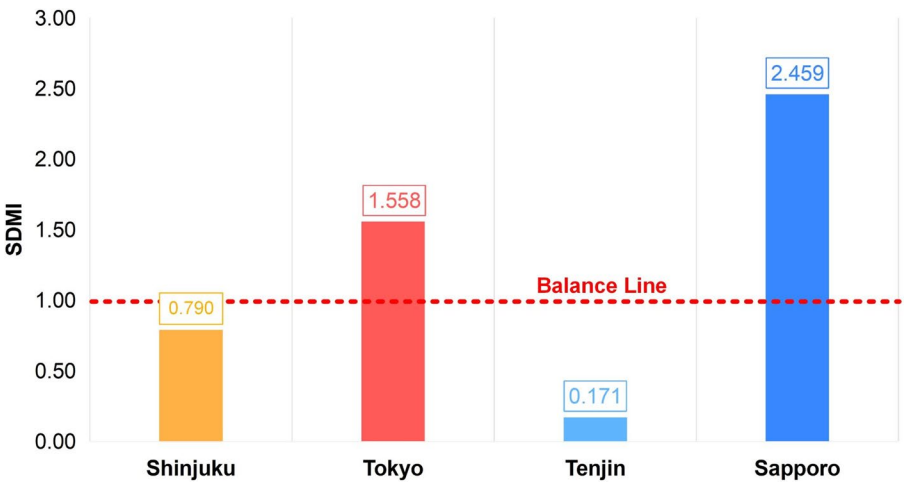



**Fig. 11** Comparative SDMI for UPSs

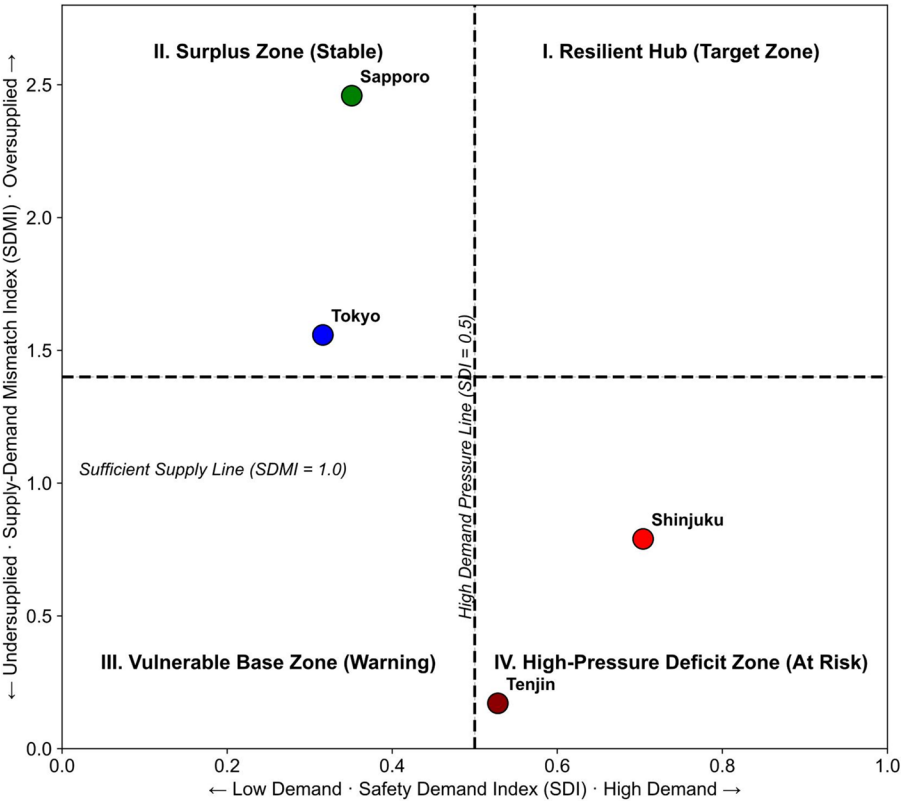



**Fig. 12** The four-quadrant strategic model for UPS safety diagnosis

defined by demand pressure (SDI) and supply adequacy (SDMI). The SDI threshold of 0.5 and the SDMI threshold of 1.0 were used to classify the cases into four diagnostic zones. The SDMI threshold of 1.0 represents the balance point between supply and demand, while the SDI threshold of 0.5 provides an operational distinction between lower and higher baseline demand pressure within the normalized index range.

Sapporo and Tokyo are located in the surplus zone, where supply adequacy exceeds baseline demand pressure. Sapporo represents a high-surplus case, supported by strong network-based safety-support capacity under moderate demand conditions. Tokyo represents a more balanced surplus case, with relatively low baseline demand and moderate safety-support supply.

Shinjuku and Tenjin are located in the high-pressure deficit zone under the baseline classification. Shinjuku represents a saturation-type mismatch: its SSI is moderate, but the demand burden is high, resulting in insufficient relative adequacy. Tenjin represents a supply-deficit type: its SSI is substantially lower than its demand level, indicating limited network-based safety-support capacity relative to structural demand.

No case falls into the high-demand surplus zone or the low-demand deficit zone. This result should not be interpreted as evidence that these typologies are absent in

Japanese UPSs more broadly. Rather, it reflects the distribution of the four selected typological cases and should be tested in larger samples.

## Equity Diagnosis of UPS Safety

### Distributional Equity

The analysis of distributional equity is conducted using a group-adapted supply model, in which the effective safety provision, adjusted for accessibility and usability by distinct demographic groups, is evaluated via the Gini coefficient. Lorenz curves (Fig. 13) demonstrate considerable inter-site variability.

The distributional equity analysis uses the group-adjusted supply model to evaluate whether effective safety-support capacity is evenly accessible across demographic

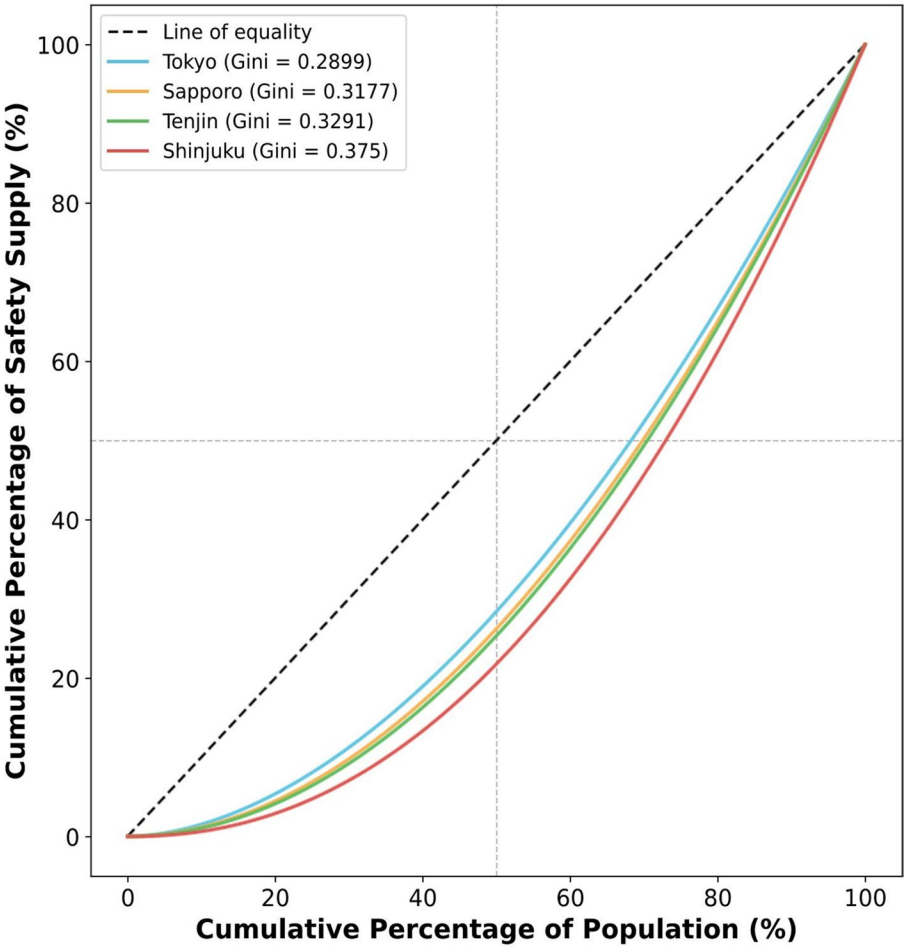



**Fig. 13** Lorenz curves of group-adapted UPS supply equity

groups. The Lorenz curves and Gini coefficients reveal clear inter-case variation (Fig. 13).

Among the cases, Shinjuku records the highest inequality level (Gini = 0.375), suggesting a relatively uneven distribution of effective safety-support capacity across groups. Tenjin also shows a high level of inequality (Gini = 0.329). By contrast, Sapporo (Gini = 0.318) and Tokyo (Gini = 0.290) show relatively more balanced distributions. These results suggest that large or functionally intensive UPSs may exhibit group-level accessibility disparities when route complexity, vertical depth, or entrance-level egress difficulty affects user groups differently.

### Needs-Response Adequacy

Equitable distribution alone does not guarantee sufficient protection if the total level of safety-support supply is inadequate relative to socially weighted demand. Therefore, the effective supply index was compared with the SDEI, which incorporates vulnerable-group exposure into the demand calculation. The SSI/SDEI ratio indicates the extent to which each UPS meets equity-sensitive demand (Table 7).

Sapporo shows the strongest needs-response adequacy, with an SSI/SDEI ratio of 1.459. Tokyo also shows surplus adequacy, with a ratio of 1.258. Shinjuku records a marginal surplus ratio of 1.024, suggesting that its safety-support supply is close to socially weighted demand and may have limited buffer under higher stress conditions. Tenjin shows a severe shortfall, with a ratio of 0.236, indicating that its network-based safety-support capacity is insufficient relative to equity-sensitive demand.

### Composite Equity Diagnosis

To synthesize distributional fairness and needs-response adequacy, the GEI was calculated for each case (Table 8). The baseline GEI calculation process is provided in Appendix F, Table F5. The GEI should be interpreted as an indicative composite diagnostic measure rather than a precise measurement of actual evacuation accessibility, because it incorporates DPR-based group adjustment and normative weighting parameters.

**Table 7** Comparison of standard vs. socially-weighted demand and adequacy

| Station | Vulnerable Pop. ratio | SSI | SDI | SDEI | SSI - SDEI | SSI/SDEI |
|---|---|---|---|---|---|---|
| Shinjuku | 0.310 | 0.556 | 0.704 | 0.543 | +0.013 | 1.024 |
| Tokyo | 0.332 | 0.492 | 0.316 | 0.391 | +0.101 | 1.258 |
| Tenjin | 0.300 | 0.091 | 0.528 | 0.383 | -0.293 | 0.236 |
| Sapporo | 0.352 | 0.864 | 0.351 | 0.592 | +0.272 | 1.459 |

**Table 8** Calculation of the GEI

| Station | Gini | 1 - Gini | SSI | SDEI | SSI / SDEI | GEI |
|---|---|---|---|---|---|---|
| Shinjuku | 0.375 | 0.625 | 0.556 | 0.543 | 1.024 | 0.824 |
| Tokyo | 0.290 | 0.710 | 0.492 | 0.391 | 1.258 | 0.984 |
| Tenjin | 0.329 | 0.671 | 0.091 | 0.383 | 0.236 | 0.453 |
| Sapporo | 0.318 | 0.682 | 0.864 | 0.592 | 1.459 | 1.071 |

Sapporo records the highest GEI value (1.071), exceeding the equity threshold of 1.0. This result reflects both strong needs-response adequacy and relatively balanced group-adjusted distribution. Tokyo records a near-threshold value (GEI = 0.984), suggesting a generally favorable but threshold-sensitive equity profile.

Shinjuku (GEI = 0.824) remains below the equity threshold. Its deficit is mainly associated with distributional inequality, despite its near-balanced adequacy ratio. Tenjin records the lowest GEI value (0.453), indicating a combined deficit in supply adequacy and composite equity performance. Its low GEI is primarily attributable to the severe gap between SSI and SDEI.

To test whether the composite equity diagnosis is sensitive to the DPR proxy, we recalculated GEI under alternative DPR-related assumptions, including relaxed and stricter HC thresholds and different elderly penalty exponents. As shown in Appendix F, Table F1, the main typological interpretation remains stable under alternative DPR-related assumptions: Shinjuku and Tenjin remain below the equity threshold, while Sapporo remains above it. Tokyo is closest to the threshold and should therefore be interpreted more cautiously. This indicates that the overall equity diagnosis is not solely an artifact of the baseline DPR classification.

## Discussion

### Built-Environment Risk as a Contextual Condition Associated with Safety-Support Capacity

The exploratory association analysis indicates that built-environment risk can be interpreted as a contextual condition related to network-based safety-support capacity. However, because the analysis is based on four typological cases, this relationship should not be interpreted as statistically generalizable or causal. Rather, it indicates that UPSs embedded in dense, complex, and weakly connected surface environments may face greater design difficulty in achieving high levels of redundancy, accessibility, and spatial robustness.

Tenjin provides an illustrative case of this association. Despite its high functional intensity, its compact and complex built-environment context is associated with the lowest SSI score among the four cases (SSI = 0.091). This pattern is consistent with disaster risk theory, which emphasizes that exposure and vulnerability should be addressed upstream rather than relying solely on compensatory downstream measures (Travieso Bello et al., 2023). The negative descriptive association between the number of refuge facilities and SSI should also be interpreted cautiously. A higher number of refuge facilities may reflect pre-existing risk recognition or compensatory provision in high-risk areas, rather than effective reduction of risk through better network-based safety-support capacity.

### Governance Context as an Interpretive Lens for Supply-Demand Imbalance

The observed supply-demand patterns are consistent with different governance and development contexts, but they should not be interpreted as definitive evidence of gover-

nance causality. The comparison suggests that UPSs shaped by stronger public planning, climate adaptation needs, and redundancy-oriented development may be more likely to show surplus or balanced safety-support capacity. By contrast, systems developed through fragmented or commercially intensive processes may face greater difficulty in aligning safety-support resources with socially weighted demand. These interpretations should be treated as hypotheses for future large-sample, longitudinal, or mixed-method research.

Shinjuku and Tenjin illustrate two different forms of potential mismatch. Shinjuku represents a demand-overload pattern: its SSI is moderate (0.556), but its SDI is the highest among the four cases (0.704), resulting in insufficient relative adequacy. This pattern is consistent with a highly commercialized and intensely used intermodal environment, where the pressure of passenger throughput and activity concentration may exceed the relative capacity of safety-support provision. Tenjin, by contrast, represents a supply-deficit pattern. Its low SSI (0.091) and SDMI (0.17) suggest that network-based safety-support capacity is insufficient relative to structural demand, possibly reflecting the constraints of fragmented development, compact commercial form, and limited redundancy.

Sapporo and Tokyo provide contrasting cases of relative surplus or near-balance. Sapporo's high SDMI (2.46) may be associated with cold-climate pedestrian protection, redundancy-oriented planning, and the role of underground space as winter mobility infrastructure. Tokyo shows a more balanced profile, possibly reflecting the coordination demands of a nationally significant transport and commercial hub. These interpretations do not establish causal effects of governance regimes, but they help generate a typological understanding of how different development contexts may be associated with supply-demand configurations in UPS safety.

Overall, the governance discussion should be understood as an interpretive lens rather than a causal explanation. The results point to the need for more systematic comparative research on how institutional coordination, ownership structure, planning regulation, and public–private redevelopment arrangements influence the distribution of underground safety-support resources.

### Infrastructure Design, Group Vulnerability, and Constructed Inequity

The group-weighted equity results suggest that infrastructure design may contribute to constructed vulnerability when physical configurations do not adequately accommodate differentiated user capacities. Critical urban theory emphasizes that vulnerability is not only an inherent personal condition, but also a relational outcome shaped by built environments, design standards, and governance practices (Ji et al., 2024; Wan & Titheridge, 2024).

Shinjuku provides an illustrative case. Its Gini coefficient of 0.375 indicates the highest level of group-differentiated inequality among the four cases. This result suggests that complex vertical transitions, long routes, and concentrated network pressure may disproportionately reduce effective safety-support accessibility for elderly users and children. Rather than proving a direct design failure, this finding indicates a potential form of recognitional inequity, in which the needs of users with different mobility and cognitive capacities are insufficiently reflected in safety-support provision.

The SDEI and GEI help make this issue analytically visible by shifting the evaluation from average safety-support capacity to socially differentiated adequacy. This methodological shift challenges the neutrality assumption embedded in many efficiency-oriented infrastructure assessments, where all pedestrians are implicitly treated as equally mobile and equally capable. The results suggest that equity-sensitive evaluation can reveal deficits that may remain hidden in aggregate supply measures. In this sense, the framework contributes to a broader understanding of urban justice in subterranean environments, while still requiring further validation through field audits, behavioral observations, and user perception surveys.

### Contribution, Limitations and Future Research Directions

This study contributes to disaster risk reduction and urban safety governance in subterranean environments by developing an integrated supply–demand–equity framework. Unlike efficiency-oriented safety assessments that primarily emphasize average evacuation time, throughput, or aggregate accessibility, the proposed framework asks for whom safety-support capacity is sufficient and where resources are misaligned with socially differentiated demand. The SSI, SDI, SDMI, SDEI, and GEI together provide a diagnostic sequence for identifying supply adequacy, demand pressure, spatial mismatch, and group-sensitive equity deficits.

Methodologically, the study translates the abstract concept of spatial justice into operational indicators for UPS safety assessment. Practically, the results suggest that underground safety governance should move beyond aggregate capacity expansion and pay closer attention to the distribution of egress redundancy, vertical accessibility, and safety-support resources for vulnerable groups. The framework may also provide a useful analytical reference for other high-density cities where underground pedestrian systems are expanding alongside transit-oriented and commercially intensive development. However, transfer to other contexts should be tested through additional case studies and locally calibrated data.

Several limitations should be acknowledged. First, this study evaluates generic network-based safety-support capacity rather than hazard-specific engineering safety performance. The SSI captures access, egress, redundancy, vertical accessibility, and spatial robustness, but it does not directly measure fire safety, flood resistance, seismic performance, ventilation, drainage, smoke extraction, or emergency power. Therefore, future applications should develop hazard-specific modules that extend the current framework (Table 9).

Second, the DPR-based assessment should be interpreted with caution. DPR is an ordinal proxy derived from vertical depth and horizontal path complexity, rather than a directly observed measure of evacuation time, barrier-free accessibility, or user-perceived difficulty. Although official station maps, GIS-measured path lengths, documented transfer information, and manual map checks were used to improve classification transparency, these procedures cannot replace field-based validation. Future research should validate and refine DPR using on-site accessibility audits, evacuation-time observations, pedestrian experiments, user perception surveys, and detailed architectural data on elevators, escalators, ramps, platform depth, and emergency facilities.

**Table 9** Hazard-specific extensions beyond the current network-based SSI

| Hazard type | Additional indicators required | Why current SSI is insufficient |
|---|---|---|
| Fire | Smoke extraction, fire compartmentation, sprinkler coverage, emergency lighting | Network redundancy cannot represent smoke spread or fire-control capacity |
| Flooding | Entrance elevation, flood barriers, drainage capacity, watertight doors | Accessibility and redundancy do not directly represent flood resistance |
| Earthquake | Structural integrity, falling-object risk, emergency power, post-earthquake accessibility | Network access does not represent seismic performance |
| Extreme heat / weather | Ventilation, thermal comfort, refuge cooling capacity | Structural demand varies temporally and physiologically |

Third, the analysis remains primarily static. Although demand amplification scenarios were introduced to test the sensitivity of SDMI under plausible stress conditions, the study does not directly model real-time pedestrian movement, peak-hour crowding, emergency behavior, or weather-related demand shifts. Future studies should integrate agent-based evacuation modelling, smartcard data, mobile signaling data, pedestrian tracking, and behavioral surveys to develop dynamic and spatiotemporal equity assessment models.

Fourth, the interpretation of governance and structural inequality should be understood within the limits of a four-case typological comparison. The empirical findings do not establish causal relationships between governance regimes and UPS safety-equity outcomes. Instead, they indicate patterns that are consistent with different development contexts and institutional arrangements. Future research should test these hypotheses using larger samples, longitudinal designs, and mixed-method evidence, including policy document analysis, stakeholder interviews, field audits, and operational data.

Finally, the robustness checks indicate that the main typological interpretation is generally stable, but not all cases are equally robust. Tenjin remains a severe deficit case, Shinjuku remains a high-demand deficit case, and Sapporo remains a surplus case under the available classification-stability checks. Tokyo is the most threshold-sensitive case because its GEI is close to 1.0. Therefore, Tokyo's equity status should be interpreted cautiously.

## Conclusion

This study developed an integrated supply-demand-equity framework for assessing network-based safety-support capacity and distributive equity in underground pedestrian systems. By applying the framework to four Japanese UPS cases, the study demonstrates its diagnostic potential across contrasting typological contexts. The empirical results should be interpreted as exploratory comparative evidence rather than statistically generalizable or causal findings.

The results indicate clear differences in supply adequacy and equity performance among the four cases. Sapporo and Tokyo show relatively higher supply adequacy under the baseline formulation, whereas Shinjuku and Tenjin present different forms of deficit. Shinjuku is characterized by high structural demand pressure and potential distributional concerns, while Tenjin shows a pronounced adequacy deficit. These patterns are consistent with differences in spatial morphology, development intensity, and governance context, but further large-sample, longitudinal, and mixed-method research is needed to test these relationships systematically.

Methodologically, the main contribution of this study lies in translating equity-oriented safety assessment into a measurable diagnostic procedure. The framework does not claim to replace hazard-specific engineering safety audits, evacuation simulations, or field-based accessibility assessments. Instead, it provides a spatial-equity lens for identifying where network-based safety-support capacity may be insufficient, where socially weighted demand is concentrated, and where group-sensitive safety deficits may require further investigation.

Practically, the findings suggest that underground infrastructure governance should move beyond aggregate capacity expansion and incorporate equity-sensitive criteria into planning, retrofitting, and risk management. In particular, indicators such as SDMI, SDEI, and GEI can help identify high-deficit and high-vulnerability nodes that may require priority attention. Rather than treating all users as homogeneous pedestrians, future UPS planning should consider differentiated mobility capacities, vertical accessibility constraints, route redundancy, and the needs of vulnerable groups in both routine and emergency conditions.

For high-density cities experiencing transit-oriented underground development, especially in East Asia, the proposed framework may provide a useful analytical reference for integrating justice and resilience into urban risk governance. However, its transferability should be tested through additional cases, locally calibrated data, and hazard-specific extensions. Future research should further develop the framework by incorporating dynamic pedestrian modelling, real-time mobility data, behavioral surveys, field audits, and participatory evaluation, so as to better capture temporal, embodied, and context-specific dimensions of vulnerability in underground pedestrian environments.

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## Declarations

**Competing Interests** The authors declare no competing interests.

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